



Neck Triangles & Deep Cervical Fascia — MCQ Practice

Anatomy GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A-E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Q Questions

1. What forms the lateral boundary of the anterior triangle of the neck?

EASY

- A. Anterior border of trapezius
- B. Anterior border of sternocleidomastoid
- C. Midline of the neck
- D. Posterior belly of digastric
- E. Middle third of clavicle

2. Which structure forms the base (inferior border) of the posterior triangle?

EASY

- A. Middle third of the clavicle
- B. Hyoid bone
- C. First rib
- D. Inferior belly of omohyoid
- E. Manubrium

3. Which fascia forms the floor of the posterior triangle?

EASY

- A. Investing fascia
- B. Pretracheal fascia
- C. Superficial fascia
- D. Prevertebral fascia
- E. Buccopharyngeal fascia

4. What is the action of one sternocleidomastoid contracting alone?

EASY

- A. Turns the face to the same side
- B. Elevates the scapula
- C. Tilts the head to the same side and turns the face to the opposite side
- D. Depresses the hyoid
- E. Flexes the neck straight forward only

5. After an excisional lymph node biopsy in the posterior triangle, a patient cannot shrug the shoulder on that side. Which nerve crosses this triangle and was likely cut?

MEDIUM

- A. Great auricular nerve
- B. Phrenic nerve
- C. Suprascapular nerve
- D. Hypoglossal nerve
- E. Spinal root of the accessory nerve (XI)

6. During carotid endarterectomy the surgeon works in the triangle bounded by the superior belly of omohyoid, posterior belly of digastric and anterior border of SCM. Which nerve must he protect here?

MEDIUM

- A. Hypoglossal nerve (XII)
- B. Phrenic nerve
- C. Spinal accessory nerve
- D. Suprascapular nerve
- E. Lesser occipital nerve

7. Which muscle divides the posterior triangle into occipital and supraclavicular triangles?

MEDIUM

- A. Posterior belly of digastric
- B. Scalenus anterior
- C. Levator scapulae
- D. Inferior belly of omohyoid
- E. Stylohyoid

8. A patient has a skin infection of the chin. Which nodes, lying in the triangle bounded by the hyoid, the anterior belly of digastric and the midline, enlarge?

MEDIUM

- A. Submandibular nodes
- B. Submental nodes
- C. Occipital nodes
- D. Supraclavicular nodes
- E. Deep cervical nodes

9. A child with rubella has enlarged nodes at the apex of the posterior triangle near the back of the scalp. Which group is this?

MEDIUM

- A. Occipital nodes
- B. Submental nodes
- C. Supraclavicular nodes
- D. Submandibular nodes
- E. Pretracheal nodes

10. Which structure pierces the investing layer of deep cervical fascia?

MEDIUM

- A. Internal jugular vein
- B. Vagus nerve
- C. External jugular vein
- D. Common carotid artery
- E. Phrenic nerve

11. The visceral layer of pretracheal fascia surrounds which group of structures?

MEDIUM

- A. Thyroid gland, trachea/larynx and esophagus
- B. Carotid artery, IJV and vagus
- C. SCM and trapezius
- D. Vertebral column and longus colli
- E. Parotid gland only

12. Which nerve supplies the posterior belly of digastric and the stylohyoid?

MEDIUM

- A. Nerve to mylohyoid (V3)
- B. Ansa cervicalis
- C. Hypoglossal nerve
- D. Glossopharyngeal nerve
- E. Facial nerve (VII)

13. Which infrahyoid muscle is NOT supplied by the ansa cervicalis?

MEDIUM

- A. Sternohyoid
- B. Omohyoid
- C. Sternothyroid
- D. Thyrohyoid
- E. Superior belly of omohyoid

14. A swelling in the submandibular (digastric) triangle is examined. Which structure is NOT a content of this triangle?

MEDIUM

- A. Submandibular gland
- B. Facial artery
- C. Hypoglossal nerve
- D. Thyroid gland
- E. Mylohyoid nerve

15. A thyroid nodule lies in the triangle bounded by the hyoid above, the midline medially, and the superior belly of omohyoid with the anterior border of SCM laterally. Which muscles lie in this triangle?

HARD

- A. Digastric and stylohyoid
- B. Thyrohyoid, sternothyroid and sternohyoid
- C. Scalenus anterior and medius
- D. Splenius and levator scapulae
- E. Mylohyoid and geniohyoid

16. A deep infection in the neck spreads down into the axilla along the subclavian vessels and brachial plexus. Which fascia's prolongation provides this route?

HARD

- A. Investing fascia
- B. Pretracheal fascia
- C. Prevertebral fascia (axillary sheath)
- D. Carotid sheath
- E. Parotid fascia

17. An abscess in the retropharyngeal space tracks downward. According to the slide figure, to which level does this space extend?

HARD

- A. Hyoid bone
- B. C6
- C. Sternal angle only
- D. Diaphragm
- E. About the 3rd thoracic level

18. Which structures make up the floor of the posterior triangle beneath the prevertebral fascia?

HARD

- A. Thyroid gland and trachea
- B. Semispinalis, splenius, levator scapulae and scalenus medius, with the subclavian artery and brachial plexus trunks
- C. Mylohyoid and hyoglossus
- D. SCM and trapezius
- E. Carotid sheath and its contents

19. According to the slides, which branch of the external carotid artery is NOT found in the carotid triangle?

HARD

- A. Superior thyroid artery
- B. Lingual artery
- C. Facial artery
- D. Occipital artery
- E. Maxillary artery

20. Which statement about the investing layer of deep cervical fascia is correct?

HARD

- A.** It encloses the thyroid gland
- B.** It forms the axillary sheath
- C.** It splits to enclose SCM, trapezius and the parotid gland, and forms the roof of the posterior triangle
- D.** It lies in front of the vertebral bodies only
- E.** It is pierced by the internal jugular vein

✓ Answer key

Q	Answer	Difficulty	Why
1	B	EASY	The anterior triangle is bounded by the inferior border of the mandible above, the midline medially and the anterior border of SCM laterally.
2	A	EASY	The posterior triangle is bounded by SCM in front, trapezius behind and the middle third of the clavicle below.
3	D	EASY	The roof is the investing fascia with platysma and the EJV; the floor is the prevertebral fascia covering the muscles.
4	C	EASY	One SCM draws the head toward the same side and turns the face to the opposite side; both together draw the head backward (per slides).
5	E	MEDIUM	The spinal root of XI crosses the posterior triangle. The slides give its surface marking from about one-third down the posterior border of SCM to one-third up the anterior border of trapezius.
6	A	MEDIUM	This is the carotid triangle, which contains the carotid arteries, IJV, vagus, hypoglossal nerve and part of the sympathetic trunk.
7	D	MEDIUM	The inferior belly of omohyoid crosses the posterior triangle: occipital above, supraclavicular (omoclavicular) below.
8	B	MEDIUM	This is the submental triangle; its contents are the anterior jugular veins and submental lymph nodes.
9	A	MEDIUM	The slides note occipital nodes in the posterior triangle enlarge in rubella and scalp infections; supraclavicular nodes are part of the deep chain.
10	C	MEDIUM	The investing fascia is pierced by the external and anterior jugular veins and the cutaneous branches of the cervical plexus.
11	A	MEDIUM	The pretracheal fascia has a muscular layer (infrahyoids) and a visceral layer around the thyroid, trachea/larynx and esophagus.
12	E	MEDIUM	Posterior belly of digastric and stylohyoid are supplied by the facial nerve; the anterior belly by the nerve to mylohyoid from V3.
13	D	MEDIUM	Thyrohyoid (and geniohyoid) are supplied by C1 fibres carried along the hypoglossal nerve; sternohyoid, omohyoid and sternothyroid by the ansa cervicalis (C1–C3).
14	D	MEDIUM	The submandibular triangle contains the submandibular gland and nodes, caudal parotid, facial, submental and lingual vessels, and the mylohyoid and hypoglossal nerves. The thyroid is in the muscular triangle.
15	B	HARD	This is the muscular (omotracheal) triangle: thyrohyoid, sternothyroid, sternohyoid, superior and inferior thyroid arteries, anterior jugular veins, and the thyroid, parathyroids, larynx, trachea and esophagus.
16	C	HARD	The prevertebral fascia is prolonged over the subclavian vessels and brachial plexus as the axillary sheath.
17	E	HARD	The slide's sagittal diagram shows the retropharyngeal space between the buccopharyngeal and prevertebral layers, reaching about the 3rd thoracic level.
18	B	HARD	The floor is the prevertebral fascia over semispinalis capitis, splenius capitis, levator scapulae and scalenus medius, with the subclavian artery and brachial plexus trunks.
19	E	HARD	The carotid triangle contains the ECA and its branches except the maxillary, superficial temporal and posterior auricular arteries.
20	C	HARD	The investing fascia surrounds the neck, splits around SCM, trapezius and the parotid (parotid fascia to the zygomatic arch) and roofs the posterior triangle.